Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/24/2013
PI Name:	Asaithamby, Aroumougame Ph.D.		
Project Title:	Mechanism of Clustered DNA Double-strand Break Repair in Response to HZE Particles in Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7201	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chen, David (The University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX13AD57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	DNA damage and consequent mutations initiate the multi-step carcinogenic process. Substantial evidence indicates that high charge and energy [HZE] particles, component of galactic cosmic rays, induced clustered DNA damages are difficult to repair, and in some instances are irreparable; this has been associated with the increased relative biological effectiveness of chromosomal aberrations, mutagenesis, and carcinogenesis. However, it is unclear why clustered lesions are difficult to repair, and the mechanisms by which unrepaired clustered DNA lesions trigger carcinogenic events are not known. The goals of this proposed research are to utilize our unique capability to quantitatively estimate the ability of cells to repair of clustered DNA lesions in response to a spectrum of HZE particles, and to dissect the genetic factors that may potentially be involved in the processing of complex DNA lesions. The specific aims of the proposed research are to test the hypothesis that: (1) the cellular ability to repair HZE particles induced clustered DNA damage is modulated by radiation quality and cell cycle stages, utilizing our newly established in situ and imaging technology; (2) homologous recombination plays a key function in facilitating the repair of double-strand breaks associated with clustered DNA lesions, using key homologous recombination factors deficient cells; (3) Fanconi Anemia (FA) complex may orchestrate the repair of complex double strand breaks (DSBs) induced by HZE particles, using a panel of FA pathway deficient human cells; and (4) DNA end-processing nucleases, such as Werner Syndrome protein (WRN) and Artemis, may facilitate the processing of complex DNA ends generated by HZE particles using human cells that are either deficient in WRN and Artemis or expressing different nuclease mutant forms of WRN and Artemis. Understanding the mechanism underlying the repair of DNA damage induced by HZE particles will have important implications for estimating the risks to Astronaut health associated with GCR exposures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the mechanism underlying the repair of DNA lesions induced by HZE particles will have important implications for estimating the risks to Astronaut's health associated with GCR exposures. Our research would also lead to the development of counter-measures to enhance astronaut’s ability to sustain HZE particles exposures in space.
Task Progress:	Summary of the progress made during the first year of this project: We established the molecular mechanism of Fanconi anemia pathway mediated suppresion of genomic instability in response to Fe particles irradiation. We found that FANCD2 is recruited to the sites of DSBs induced by Fe particles in S/G2 phase cells. FANCD2 is critical for cellular sensitivity, clustered DSBs repair, and replication forks recovery in response to Fe irradiation. In addition, we found that FANCD2 stabilizes RAD51 at the sites of DNA lesions induced by Fe particles and this is critical for the repair of DSBs. Further, the unrepaired DSBs result in the formation of higher numbers of chromosomal aberrations and finally lead to the cellular transformation. Thus, our results clearly reveal the significance of FA pathway in the suppression of genomic instability in human cells following space radiation exposure.
Bibliography Type:	Description: (Last Updated: 03/12/2018)
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Cellular response to HZE particles irradiation is dependent on cell culture models." International Symposium on Radiation Science 2013 (Proton and heavy ion effects in relationship to risk and cancer treatment) and 24th Annual NASA Space Radiation Health Investigators’ workshop satellite meeting, Taipei, Taiwan, May 20-21, 2013. Oral presentation. International Symposium on Radiation Science 2013 (Proton and heavy ion effects in relationship to risk and cancer treatment) and 24th Annual NASA Space Radiation Health Investigators’ workshop satellite meeting, Taipei, Taiwan, May 20-21, 2013. http://homepage.ntu.edu.tw/~ntuylbme/ISRS2013/program.html , May-2013
Abstracts for Journals and Proceedings	Chen DJ, Asaithamby A. "Repair of HZE particles induced DNA double-strand breaks repair in human cells." International Symposium on Radiation Science 2013 (Proton and Heavy ion effects in relationship to risk and cancer treatment) and 24th Annual NASA Space Radiation Health Investigators’ workshop satellite meeting, Taipei, Taiwan, May 20-21, 2013. Oral presentation. International Symposium on Radiation Science 2013 (Proton and heavy ion effects in relationship to risk and cancer treatment) and 24th Annual NASA Space Radiation Health Investigators’ workshop satellite meeting, Taipei, Taiwan, May 20-21, 2013. http://homepage.ntu.edu.tw/~ntuylbme/ISRS2013/program.html , May-2013
Abstracts for Journals and Proceedings	Zhu J, Su F, Gerelchuluun A, Chen DJ, Asaithamby A. "Fanconi anemia pathway function in genome stability maintenance in response to HZE particles irradiation." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. Oral presentation. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Abstracts for Journals and Proceedings	Gerelchuluun A, Zhu J, Su F, Chen DJ, Asaithamby A, Tsuboi K. "Homologous recombination pathway plays a major role in high-LET radiation induced DNA double-strand break repair." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. Poster Presentation. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Abstracts for Journals and Proceedings	Ye Z, Cox B, Asaithamby A, Chen DJ, Wu H. "Chromatin folding, fragile sites, and chromosome aberrations induced by low- and high-LET radiation." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. Poster presentation. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013